

HuffPost News Category Classification with DistilBERT

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Introduction and Problem Definition:

Problem description and motivation: picked HuffPost as my final project. The work involves multiclass classification for text dataset problems. The objective is to implement a way to predict a category using an article's headline and short description, which 34 categories it should fit into. The goal is to have news article be placed into one of 34 selected topics.

This problem topic is useful when applied to handle large quantities of news content, particularly in online format. It is not practical to manually sort massive amounts of news articles for categorical use. Using a classifier used in this project is a good approach to sorting incoming articles to place in the right section. It would be good for organizational purposes, searches, and what to recommend to readers.

This topic is meaningful because category classification impacts finding articles for readers, which in turn would impact how many people might read certain articles based on search capabilities. Misclassification of an article could lead to less people reading it because it may not come up in searches; for example, something in style could be classified as education because it could be related to education in fashion. As such, an appropriate method to classify goes beyond organization of news content; it lends itself to improved readership experience. Consequently, newspaper companies can support their efforts to be more efficient way of classifying and organizing their content as well by having a news platform that uses the model to sort out large amount of articles for search, recommendation, in addition to organizational efforts: a difficult task given that headlines are very short with categories that overlap.

Project goals in both business and data-science terms:

Business goals: The business goal is to facilitate a news platform to identify articles with correct topic, when they are added to their system. This is quite efficient, given that the articles would not have to be sorted by a person manually, a time-consuming process and more error prone when potentially massive amounts of articles could be involved in this. As mentioned, organization is key for better user experience for searches and recommendations as well. Thus, the model is good for initial look for classification, but for the articles that fall under grey area for modeling, a human being can categorize it to help with classification.

Data science goals: The data-science goal is to be able to categorize a news article using headline and short description by being able to predict as one of the 34 categories. Macro-F1 was used as

the primary metric because of imbalance to treat each class equally. Weighted-F1 used to show model performance overall while still looking at class size. Accuracy showed percent that is right percentage, precision showing the prediction of frequency of correct label, and recall for how many articles for each category the model can find. One of the constraints is being able to run it in Colab, with GPU and high RAM, and still needs to be able to run on reasonable amount of time in this computing environment. Other constraints include class imbalance across the 34 categories, short headlines and descriptions, and categories that overlap, such as for Wellness and Healthy Living.

Dataset description: Source of the dataset: For this project a text data was selected, namely HuffPost. Its purpose is to categorize news articles. Hugging Face was used to load the data for the data science work through khalidalt/HuffPost. The raw dataset had 200,853 articles.

Number of classes and their distribution: Originally, the data had 41 categories for topics. EDA showed that some categories were similar. Examples are PARENTS/PARENTING, STYLE & BEAUTY/STYLE, and THE WORLDPOST/WORLDPOST. Decision was made to merge these categories that are overlapping pairs, and consequently, modeling ended up using 34 categories, instead of 41 categories. In terms of class distribution, the data is highly imbalanced. The largest class is POLITICS with 32,739 articles and smallest class is EDUCATION with 1,004 articles. Given such difference, such class imbalance was one of the main concerns for this text data.

Key input modalities (text headlines for HuffPost): This project is for modeling text information as the main key input modality. Hence, HuffPost dataset does not have any images or audio for modeling purposes. Basically, it is news in text format, with article headline and short description: inputs for classification of categories.

Summary of features used: The dataset has fields, such as category, headline, short_description, authors, link, and date. Headline and short_description as the input features are used for modeling needs. I combined text fields, so the model had more context than just the headline alone. The category column was used as the target label for classification. author, link, or date fields were not used since this project is about being able to predict which topic new category belongs to, not about article text or using additional metadata.

Any challenges; imbalance (HuffPost, mislabeled data, class overlap, etc.): EDA step showed that the dataset has challenges ahead in order to do modeling. The concern that stood out the most was severe class imbalance, with some categories having a lot more than other categories. This in turn could have the model pay more attention to larger classes like POLITICS and focus less on smaller classes. Additionally, there were 6 empty headlines and 19,712 empty short descriptions, meaning some examples have less information to learn from. I also noticed 717 exact duplicate headlines, and about 2,000 near-duplicate headlines (this includes exact duplicate headlines), that can potentially lead to leakage if similar articles show up in both training and testing.

Additionally, there are class overlaps that are more natural, like BLACK VOICES and LATINO VOICES. Someone could be black and Latino at the same time and express their view and then the article could go into either category. As such, this classification is more difficult since some articles could logically be labelled for more than one category.

Brief EDA Summary: The dataset was cleaned, resulting in 200,158 articles with 34 categories after preprocessing and merging of categories were done. Interestingly, EDA did reveal imbalance in data, but the data is sufficient in size to do modeling, although there was missing text information, labels that overlap, and duplicate data. Combined text of headline and short description showed 28 words for median length for articles and 56 words for 95th percentile. This observation supported in directing sequence length for tokenization. Appendix has what is needed for table/plot of distributions for class and statistics for text length.

Methodology:

Preprocessing pipeline: To get preprocessing going, I combined headline and short_description fields as one text field. This became the input. Those without short description, however, were kept for the headline to utilize this information. After this concatenation, extra spaces were eliminated along with lowercasing the text, removed duplicates along with items that were completely empty for the rows, and those with fewer than 3 words were also removed, since they would not provide much detail for classification task ahead. These steps supported lowering noise level of this data, as well as reducing leakage that could be caused by articles that are repeats. Categories that were similar were also merged: examples include PARENTS with PARENTING, and THE WORLDPOST with WORLDPOST. Merging resulted in 34 categories instead of original 41 categories. The categories were then encoded to be numeric to do the

training step. Data split followed this 80% training/10% validation/10% test that was stratified. Stratification was thought of because of imbalanced data to give each split class distribution that is similar. While validation was implemented when training was being done, the test set itself was held back to be used in the final evaluation at the end.

The final model tokenizer, `DistilBertTokenizerFast`, combined text used for input to `input_ids` and an `attention_mask`, that is needed by DistilBERT (final model) for inputs. Maximum length of 128 token with padding and truncation was used here. These steps proved to be adequate given that concatenated text had median length of words of 28 words having 56 words for 95th percentile. Since DistilBERT has built in tokenizer and embeddings that are pretrained already, TF-IDF was not used in the final model for vectorization. The final model also has tokenization for subword to help with rare words so that they will not be treated as complete unknown. DistilBERT setup did not have in its range Keras TextVectorization layer that was used by the baseline modeling giving vocabulary size of 48,194.

In order to have smaller classes to be focused more while in training, a computation of balanced class weights was done for training labels. This meant that the model would not just spend its time paying attention to mostly to larger categories. This can be seen as COLLEGE category receiving high weight of about 5.153 even having few sample size. Appendix has class distribution figure and statistics for text-length.

Model architecture and justification: DistilBERT with `distilbert-base-uncased` was the model of choice as the final model, utilized with partial fine-tuning. This is justified given that HuffPost is text based for mostly for article headline and short description, which the model is appropriate since it is pretrained language model for the English text. DistilBERT is a smaller version of BERT, so it is better for Colab environment for training where computing could be an issue. The model uses [CLS] representation for its classification task after the text goes through the model. It goes through a pre-classifier layer that is linear, an activation function, with a dropout of 0.1. It then goes to final linear layer to have an output of 34 logits. As a transformer model, it does not use CNN blocks. Main architectural components: pretrained embeddings, DistilBERT transformer layers, drops, and lastly, classification head that is final.

I worked with models that used a simple embedding model with global average pooling, a BiGRU model, and a DistilBERT model. I really thought BiGRU would do better than the

baseline since it could use word order. Surprisingly, it did a little worse than the baseline. DistilBERT performed the best out of the models, particularly looking at macro-F1. Pretrained language representations were shown to be more important than just working with a from-scratch model. Appendix has the table of results for this. Given the outcome, the final model was set from a frozen DistilBERT to a partially fine-tuned DistilBERT architecture. Consequently, the embedding layer and the lower four transformer layers were frozen for the setting, with the top two transformer layers and the classification head trainable. This made about 14.8M out of 67.0M parameters trainable (about 22.1%). This decision was made since the lower layers could keep general English language features from pretraining, but the upper layers could perform well enough to be flexible to be able to adjust to the HuffPost topic category needs.

To sum it up, model architecture was reasonable choice and the right choice since it performed best, it showed that models learning from scratch did not really work well enough. DistilBERT outperformed for text classification, with partial fine-tuning as the next step in all this to adjust properly for the 34 categories, leaning to more efficient training and having reduced risk of overfitting.

Training strategy: Weighted CrossEntropyLoss was used for loss function for training, a weighted version of cross-entropy for multiclass classification for the task ahead. This is appropriate given that the model predicts one label out of 34 categories for each example. Weighted version was necessary because of dataset is imbalanced for categories, among the largest difference in classes being POLITICS and EDUCATION. Furthermore, class weights were calculated only from the training labels only, with sklearn's balanced class weights. This way smaller classes could have more impact while in training to deal with imbalance.

AdamW was the chosen optimizer and paired with learning rate of $2e-5$. AdamW was selected since it handles transformer tuning and weight decay well and much better choice than regular Adam in this regard. Linear learning rate schedule with 10% warmup was also used here. Having warmup helped since classification head started with random weights, even though DistilBERT layers already had pretrained weights. After warmup was done, the learning rate slowly decreased for the model to have smaller updates later in training process.

For regularization, weight decay of 0.01 and dropout of 0.1 in the classification head were utilized. These items supported in reducing overfitting but still be able for the model to

learn from the HuffPost training data. The final model was trained with a batch size of 32 for up to 10 epochs, with a maximum sequence length of 128 tokens. This was sufficient for most articles, while being computationally friendly enough in Colab setup environment.

Three callbacks were used. First one, early stopping was used to monitor validation loss with a patience of 3 epochs; this way training stopped if validation loss did not improve for three epochs in a row, which it did. Second one, the best model weights got to be saved and restored when training stopped; it meant that the final model came from the best validation-loss epoch rather than from just the last epoch. This way it is performance related. Third one, used the learning rate scheduler rather than a separate LR reduction callback. This process reduced the learning rate after warmup; it allowed for training to be more stable near the end of its run. Appendix has the table for full hyperparameter values.

Performance validation: The final model was analyzed using a split of 80/10/10 that was stratified for training, validation, and testing, respectively. Stratification was done because data is imbalanced for categories. Using this method enabled each split to have about same class distribution. Training set was used in order to fit the model, while validation set used when training was being done; lastly, test set was set aside for evaluating at the end. Validation loss and validation accuracy was looked at after each epoch during training. Additionally, validation loss was used for early stopping and checkpointing; this way the final model was from the epoch that did best on the validation data itself rather than from the last epoch. This helped so that the model is not kept for only doing better on the training set. The test set was not used for tuning, early stopping, or for choosing the final model. Instead, the test set was used once after the training was done to evaluate the final performance metrics.

In order to handle class imbalance, balanced class weights were used rather than oversampling. They were calculated from the training labels. They way it did not utilize information from validation or test sets for training. Exact duplicates were removed before splitting the data and did keep the test set separate during this whole process to help lower the chance of data leakage. This environment was appropriate to check on how good a model could generalize for unseen HuffPost articles.

Results and Evaluation:

Quantitative results (tables and plots): ** Refer to appendix for the visuals. **

Training, validation, and test accuracy: See appendix

Training and validation loss curves: See appendix

Confusion matrix for test prediction: See appendix

Dataset-specific metrics: HuffPost (e.g. macro-F1, weighted-F1, and a class frequency table):
See appendix

Interpretation of results: Which classes are easiest/hardest Results indicate that some HuffPost categories were easier for the model than others to do. Top easiest classes were STYLE, FOOD & DRINK, WEDDINGS, TRAVEL, and DIVORCE, given that they had the highest F1 scores, approximately at 0.80 or higher values. This means, basically, the final model was more consistent with these categories and made fewer mistakes compared with the lower-scoring classes. The top hardest classes were FIFTY, GOOD NEWS, WOMEN, WORLD NEWS, and IMPACT, given that they had some of the lowest F1 scores when observed. They were harder because they are broader and overlap with other categories. As an example, WOMEN or IMPACT articles could also deal with politics, so the model may not be able to distinguish some categories well from others. As such given these results, the easiest classes had clearer separation of classification done, while the hardest ones were more mixed with other HuffPost topics in terms of classification. **Patterns in the confusion matrix** Confusion matrix indicated that most mistakes were between categories that seemed related in some way. For example: There were darker spots in the POLITICS row under MEDIA, WORLDPOST, CRIME, and BLACK VOICES, so some politics articles were getting predicted as those categories instead. There were additional similar patterns for WELLNESS, where some darker off-diagonal spots were near PARENTING, FIFTY, and SCIENCE. Another area that stood out was ENTERTAINMENT, COMEDY, ARTS, and CULTURE & ARTS, where the categories seemed to mix together more than others as well. Consequently, this matrix would indicate that the model mostly has challenges when categories were close in topic from separating them. This isn't due to making random mistakes by the model. **Evidence of overfitting/underfitting** There was some overfitting as indicated by training results, but not so serious that would make the model useless. Basically, while the training loss kept going down, the validation loss did best at

epoch 5 then it started to go up. Early stopping was useful because it restored the best weight from epoch 5 rather than using what was available for later epochs. Essentially, validation accuracy was 0.6538 and the test accuracy was 0.6547, leading to indicate that the model performed almost well on test data that is unseen. So, consequently, the model would generalize well. **Impact of imbalance (HuffPost)** Class imbalance is an issue in this dataset in terms of its impact on the results. It seems that larger and clearer classes performed better in this situation, while smaller and/or overlapping classes had lower F1 scores as the analysis indicated. Some numbers include weighted-F1 had an outcome of 0.6676, but the macro-F1 was lower at 0.5845. This would lead to show that the model had greater influence where larger classes counted more, but performance was less even across all classes in this case. Additionally, the macro-F1 was still higher than the corrected baseline macro-F1 of 0.4766, and partial fine-tuning helped to obtain these numbers. To sum it up, the model did improve, but the main struggles were still class imbalance, along with overlaps between HuffPost categories without clear distinction for some categories.

Summary of final model performance: Instead of comparing multiple models, provide a concise explanation of: Why this model represents my “best” solution: The partially fine-tuned DistilBERT model was my best choice, as indicated by the outcome of experiments in this project; it had the most robust final results and still being realistic to train, given the computational limits and the time needed to train models. It went up to 0.6547 test accuracy, 0.5845 macro-F1, and 0.6676 weighted-F1. Given that HuffPost data is very imbalanced, the attention should be more on macro-F1, since it can look at each category more evenly. The macro-F1 also improved over the baseline score of 0.4766; a clear indication that partial fine-tuning did help in this situation. **What its performance indicates about the difficulty of the task:** These performance results lead to understanding that the task was difficult for the model, given that in particular the model had to decide between 34 categories using only the headline and short description, very little information. While categories were easier to separate, others overlapped a lot without clear distinction. For example, articles about politics could also connect to world news. It is comprehensible why some mistakes did occur because the labels were not always easy to separate them individually. **Any surprising behaviors or strengths/weaknesses:** One surprising performance result was with POLITICS category. It had high precision for this but lower recall; this means that when the model predicted POLITICS, it was usually correct

generally, but it still missed a lot of articles that were POLITICS, a source of mistakes. The model did better for categories like STYLE, FOOD & DRINK, WEDDINGS, TRAVEL, and DIVORCE; they had stronger F1 scores than other categories. The model struggled with categories like FIFTY, GOOD NEWS, WOMEN, WORLD NEWS, and IMPACT, given that the labels were broader or overlapped more with other topics as well. As a follow-up comment, the model improved compared with the baseline when analysis was done, but class imbalance and category overlaps were still the main weaknesses for this dataset.

Discussion and Reflection: Limitations and sources of error: Main concern of limitation was data: had very imbalanced classes, such as POLITICS having many more examples than smaller classes like EDUCATION. Balanced class weights did help with this; the smaller classes still had weak F1 scores. Also, some overlapped, such as POLITICS, WORLDPOST, and WORLD NEWS: difficult to classify correctly. Another issue was that the text was short and sometimes noisy because the model only used the headline and short description. Additionally model limitations can be seen. DistilBERT is smaller than other transformer models; fine-tuned the top layers and classification head for modeling. This made training faster and helped control overfitting; however, it limited how much the model could adjust to the given dataset. The results would be varied with different hyperparameters, such as learning rate, dropout, batch size, or early stopping. The main source of errors really stems from imbalance, categories that overlap, short text, and the limits of this model setup environment.

Potential improvements: If I had more time, I would do improvements. First, test for focal loss rather than only using weighted cross-entropy, since smaller classes, like FIFTY and GOOD NEWS, still had weak F1 scores. Focal loss could support the model to impact more on harder categories. Second, I would try a more robust transformer model like RoBERTa or DeBERTa, since DistilBERT is a small transformer and can ignore some of the small differences when categories overlap. Third, I would try to improve tokenization or embedding approach potentially with a model trained more on news text. It might help with understanding headlines and short description classification efforts better for this data. Fourth, I would improve the cleaning by working with more on near-duplicate articles. I already handled missing text and exact duplicates. Near-duplicates are harder to deal with because they may not be the same but have

similar patterns that repeat themselves. These changes could help with class imbalance, short text, and overlapping HuffPost categories.

Reflection on lessons learned: Lessons learned from this project are that having good performance was not just about using chosen, final model. Other things matter for model performance like data cleaning, handling class imbalance, and fine-tuning decisions. A strategy using partial fine-tuning supported better model outcome results, as well as class weights helping with the smaller classes. Tuning can impact results for the model because small changes like the number of unfrozen layers, learning rate, or early stopping can positively impact performance. If I did this project again, I would spend more time on EDA at the beginning. This way the setup for model evaluation codes can be done sooner to evaluate macro-F1 and per-class results after each model training run. This would make it easier to see which changes helped, particularly when it comes to the smaller classes and not just the overall accuracy view. Since some HuffPost labels overlap a lot, mistakes were understandable during the categorization process. An article could fit more than one category, which made the task harder than just training a model and expecting perfect results at the end.

Appendix:

Table Statistics and data information

Statistic	Value
Total raw articles	200,853
Categories (raw data / after merge data)	41 / 34
Largest class (POLITICS)	32,739
Smallest class (EDUCATION)	1,004
Imbalance ratio (max / min)	About 32.6 x
Empty headlines	6
Empty short description	19,712
Exact-duplicate headlines	717
Near-duplicate headlines (looked at first 30 characters)	About 2,000
Combined text: average words	29.3
Combined text: median	28
Combined text: 75th / 90th / 95th percentile	35 / 49 / 56
Combined text: max words	245
Articles < 3 words	207 (combined text)

Plot: Raw data histogram before cleaning

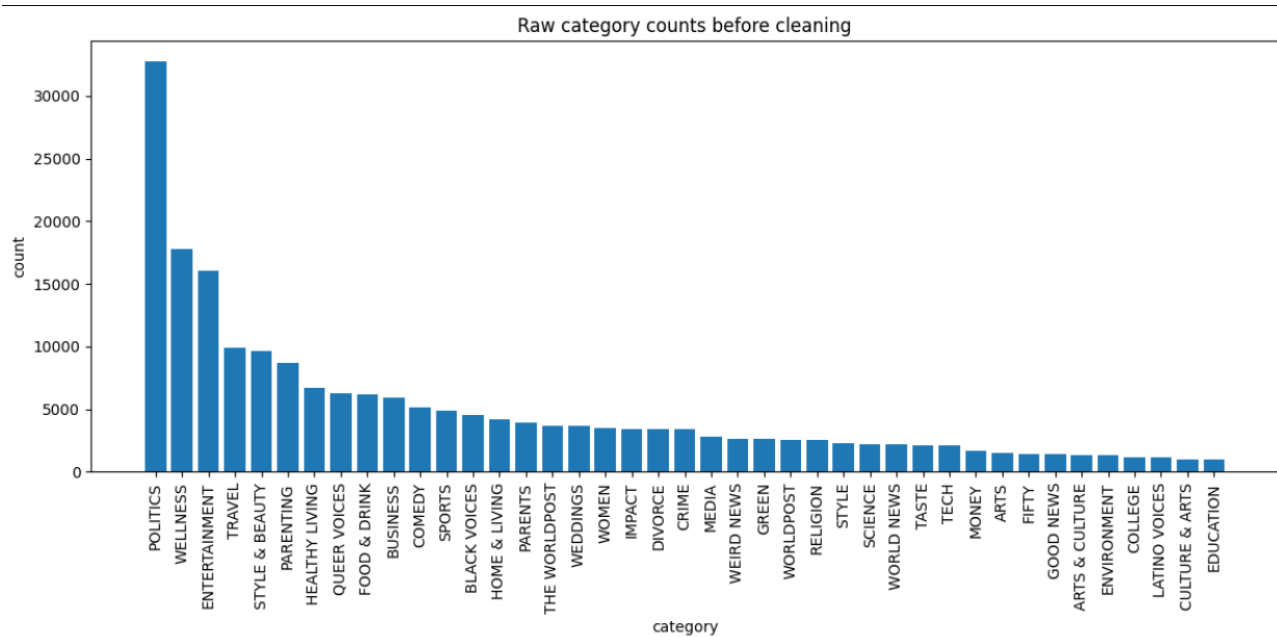


Table Category Counts: With Cleaning and Merging of Categories

34 categories and 200,158 total samples

Category	Sample number	Category	Sample number
POLITICS	32,700	DIVORCE	3,422
WELLNESS	24,490	CRIME	3,400
ENTERTAINMENT	16,048	WOMEN	3,397
PARENTING	12,546	ARTS	2,842
STYLE	11,757	MEDIA	2,811
TRAVEL	9,877	WEIRD NEWS	2,667
FOOD & DRINK	8,315	RELIGION	2,545
QUEER VOICES	6,308	SCIENCE	2,178

WORLDPOST	6,232	WORLD NEWS	2,176
BUSINESS	5,925	TECH	2,033
COMEDY	5,089	MONEY	1,706
SPORTS	4,879	GOOD NEWS	1,396
BLACK VOICES	4,521	FIFTY	1,392
HOME & LIVING	4,160	COLLEGE	1,142
ENVIRONMENT	3,937	LATINO VOICES	1,129
WEDDINGS	3,651	CULTURE & ARTS	1,029
IMPACT	3,454	EDUCATION	1,004

Table: Model results

Model	Test Acc	Macro F1	Total available Params	Train Time
Baseline (Embedding → GAP → Dense)	0.5575	0.4766	3.09M	~0.6 min
Custom (BiGRU)	0.5465	0.4714	3.14M	~1.6 min
DistilBERT (head only, frozen base)	0.6511	0.5025	~67M	~19.9 min

Table: Final DistilBERT Training Hyperparameters

Hyperparameter	Value	Notes
Optimizer	AdamW	Used most for fine-tuning transformers, so I went with it. The weight decay is built in, so I did not need a separate L2 term for this consequently.
Learning rate	2e-5	Standard starting point for fine-tuning DistilBERT. I did not want to go higher because it could mess up the pretrained weights.
Weight decay	0.01	Small amount in order to help with overfitting, handled by AdamW in this case
LR schedule	Linear with warmup	Goes up during warmup then comes back down. Pairs well with AdamW for transformers this setup
Warmup ratio	0.10	First 10% of training steps (5,004 out of 50,040). The classification head starts random while the rest is pretrained, so warming up slowly seemed safer than jumping in at full LR. So went with this setup
Loss function	Weighted CE	CrossEntropyLoss with balanced class weights. Without this, the model would mostly just learn POLITICS since it has 32 times more samples than EDUCATION. It helps with this unbalanced data problem for categories.
Batch size	32	Fit fine on the Colab A100 with max length 128. Bigger batch sizes could run into memory issues, so had to be avoided.
Max epochs	10	Upper limit. Early stopping kicked in before getting there but still need a limit.
Early stopping	patience = 3	Stops if validation loss does not improve for 3 epochs in a row. Saves time and avoids overfitting. It also saves time.

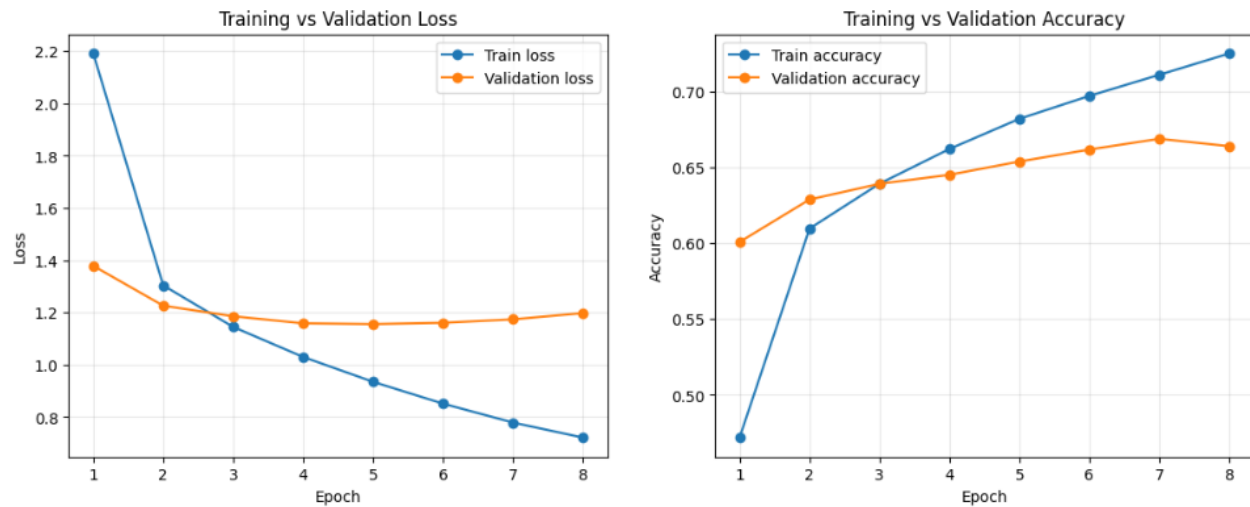
Hyperparameter	Value	Notes
Checkpointing	Best val_loss	Saves the best model weights inside EarlyStopper instead of just keeping the last epoch, a necessary step.
Dropout (head)	0.1	DistilBERT default. Left it as is since it already does some regularization
Max sequence length	128 tokens	Covers about the 95th percentile of token lengths. Additionally, any longer length is just wastes of computing resources.
Frozen layers	Embeddings + layers 0-3	Keep the general English knowledge from pretraining. No need to retrain that part, which is implemented here.
Trainable layers	Layers 4-5 + head	Top transformer layers + classification head. Around 14.8M of 67M parameters (22.1%) actually update during training.

Quantitative results (tables and plots):

Training, validation, and test accuracy:

Split for each set	Accuracy values	Comments
Training	0.6820	Best epoch (epoch 5, it restored using early stopping)
Validation	0.6538	Best epoch (epoch 5, lowest value for val_loss = 1.1556)
Test	0.6547	Basically a held-out with 10% test set, evaluated once only at the end

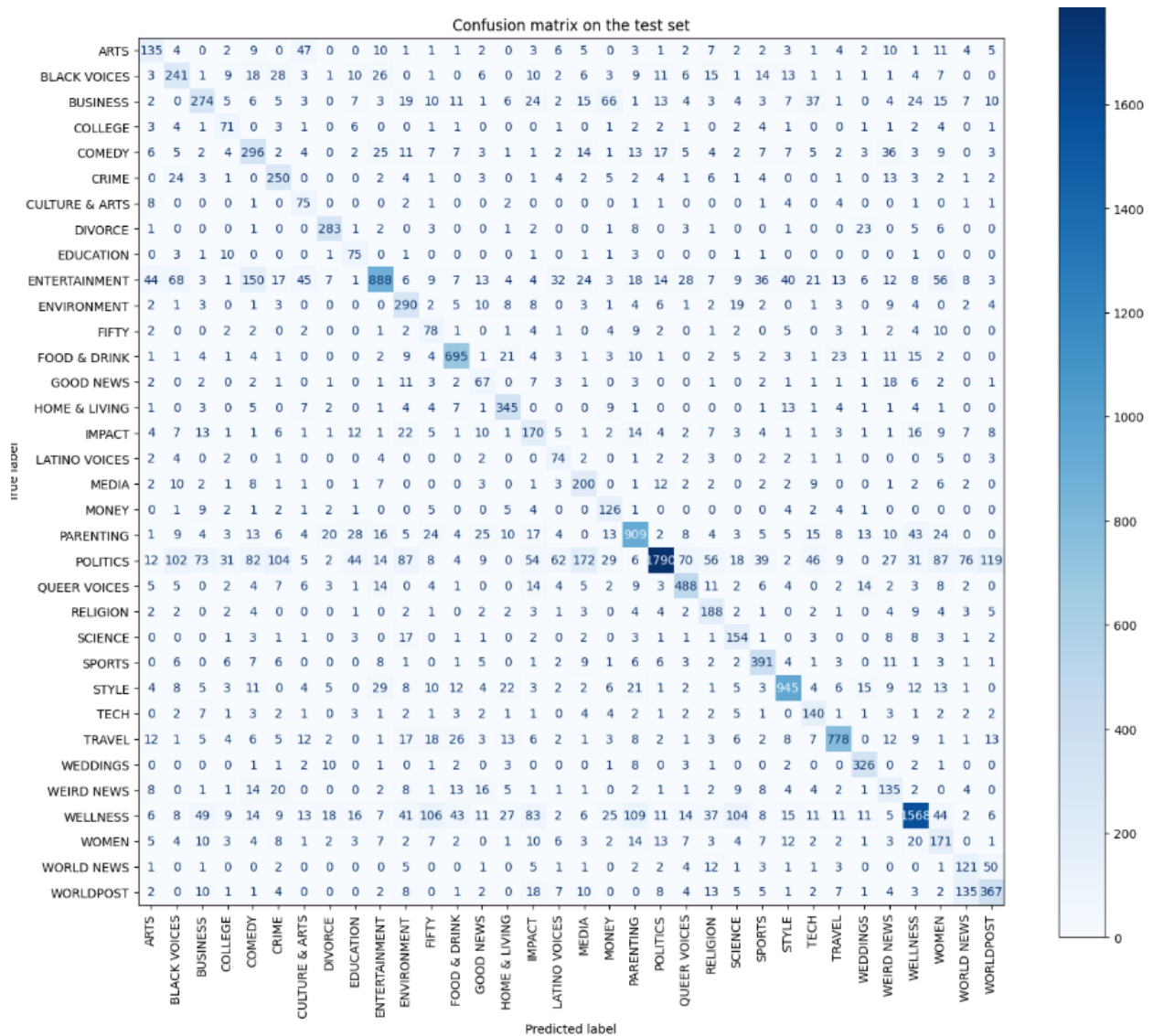
Training and validation loss curves:



Per-epoch training and validation metrics: best epoch highlighted in the table.

Epoch number	Train Loss Value	Train Acc Value	Val Loss Value	Val Acc Value
1	2.1918	0.4720	1.3786	0.6008
2	1.3038	0.6097	1.2260	0.6288
3	1.1447	0.6392	1.1856	0.6391
4	1.0297	0.6621	1.1585	0.6451
5	0.9346	0.6820	1.1556	0.6538
6	0.8509	0.6971	1.1606	0.6617
7	0.7786	0.7110	1.1737	0.6688
8	0.7209	0.7250	1.1978	0.6639

Confusion matrix for test prediction:



Dataset-specific metrics: HuffPost (e.g. macro-F1, weighted-F1, and a class frequency table):

** Final test-set metrics: DistilBERT model (partially fine-tuned approach). **

Metric value	Value	Basic result
Test Accuracy	0.6547	This is overall fraction of correct predictions of the analysis
Macro-F1	0.5845	Have each class weighted equally here (treats

Metric value	Value	Basic result
		imbalance fairly using this approach)
Weighted-F1	0.6676	Have each class weighted by support (reflects overall scale here)

Per-class and overall values for: precision, recall, F1, and support on the test set.

Category	Precision	Recall	F1-score	Support
ARTS	0.4891	0.4754	0.4821	284
BLACK VOICES	0.4635	0.5332	0.4959	452
BUSINESS	0.5638	0.4628	0.5083	592
COLLEGE	0.3966	0.6228	0.4846	114
COMEDY	0.4405	0.5815	0.5013	509
CRIME	0.5051	0.7353	0.5988	340
CULTURE & ARTS	0.3138	0.7282	0.4386	103
DIVORCE	0.7861	0.8275	0.8063	342
EDUCATION	0.3488	0.7500	0.4762	100
ENTERTAINMENT	0.8260	0.5533	0.6627	1,605
ENVIRONMENT	0.4957	0.7360	0.5924	394
FIFTY	0.2468	0.5612	0.3429	139

Category	Precision	Recall	F1-score	Support
FOOD & DRINK	0.8167	0.8363	0.8264	831
GOOD NEWS	0.3300	0.4786	0.3907	140
HOME & LIVING	0.7203	0.8293	0.7709	416
IMPACT	0.3680	0.4928	0.4213	345
LATINO VOICES	0.3190	0.6549	0.4290	113
MEDIA	0.4049	0.7117	0.5161	281
MONEY	0.4038	0.7368	0.5217	171
PARENTING	0.7531	0.7243	0.7384	1,255
POLITICS	0.9251	0.5474	0.6878	3,270
QUEER VOICES	0.7316	0.7734	0.7519	631
RELIGION	0.4736	0.7402	0.5776	254
SCIENCE	0.4151	0.7064	0.5229	218
SPORTS	0.6896	0.8012	0.7412	488
STYLE	0.8514	0.8036	0.8268	1,176
TECH	0.4375	0.6897	0.5354	203
TRAVEL	0.8635	0.7874	0.8237	988
WEDDINGS	0.7671	0.8932	0.8253	365
WEIRD NEWS	0.3824	0.5056	0.4355	267

Category	Precision	Recall	F1-score	Support
WELLNESS	0.8639	0.6403	0.7355	2,449
WOMEN	0.3360	0.5029	0.4028	340
WORLD NEWS	0.3176	0.5550	0.4040	218
WORLDPOST	0.6046	0.5891	0.5967	623
Accuracy			0.6547	20,016
Macro avg	0.5485	0.6637	0.5845	20,016
Weighted avg	0.7177	0.6547	0.6676	20,016